

Habana Almendrón Routes

History

Yasmani drives an *almendrón* (classic American car used as a shared taxi) through the streets of Havana, picking up passengers at numbered zones (1–9) and dropping them off at lettered destinations (A–Z) to earn pesos. He must reduce speed when driving through caballitos (police checkpoints). In Cuba, shared taxis pick up and drop off multiple passengers along the route rather than going directly to a destination. Help Yasmani calculate his travel statistics and earnings.



Objective

Your goal is to **maximize the total pesos earned** by selecting which passengers to serve and finding the most efficient route that respects all constraints.

The City Grid

The city is an $R \times C$ grid. Each cell has a traversal cost:

- . : Street (1 minute).
- # : Blocked (impassable).
- C : Checkpoint (3 minutes).
- T : Traffic (5 minutes).
- 1-9 and A-Z: Pickup/Dropoff zones. These cost **1 minute** (same as a street) and appear exactly once on the grid.

Movement and Taxi Rules

- **Movement:** N, E, S, W. Moving to a cell costs that cell's time.
- **Start/End:** Yasmani starts at $(START_R, START_C)$ (always a . cell) and must return there.
- **Capacity:** Max CAP passengers in the taxi at any time.
- **Patience:** Each passenger i waits at their pickup zone until their *patience* time (measured from $t = 0$). If Yasmani arrives at $t > patience_i$, the passenger is gone.

Important Clarifications

- **Patience applies only to pickup time.** Once a passenger is picked up, there is no time limit for dropoff (other than T_{MAX}).
- **Dropoff zones are always uppercase letters A–Z, excluding C and T** (which are reserved for Checkpoints and Traffic).
- **Pickup zones are digits 1–9, and multiple passengers may share the same pickup zone** (e.g., several passengers might be waiting at zone '1').
- **When arriving at a cell, dropoffs happen before pickups** to free up space first.
- **Moving cost:** The time cost is that of the **destination** cell (standard in ICPC grid problems).

Input

- Line 1: Four integers R, C, CAP, T_{MAX} .
- Line 2: Two integers $START_R, START_C$.
- Next R lines: The grid (each line has C characters).
- Next line: An integer N (number of passengers).
- Next N lines: **pickup_zone** (digit 1-9), **dropoff_zone** (uppercase letter A-Z, excluding C and T), **payment**, and **patience**.

Output

- Line 1: Maximum pesos earned (integer).
- Line 2: Number of moves M .
- Lines 3 to $M + 2$: Each move as " $r\ c$ " (destination cell coordinates).
- If no passengers can be served, output 0 and 0.

Constraints

- $5 \leq R, C \leq 100$
- $1 \leq CAP \leq 9$
- $10 \leq T_{MAX} \leq 3000$
- $1 \leq N \leq 12$
- $20 \leq payment \leq 1000$
- $5 \leq patience \leq T_{MAX}$

Scoring

- **Subtask 1 (20 pts):** $N \leq 8$ and $patience = T_{MAX}$ for all passengers.
- **Subtask 2 (30 pts):** $N \leq 12$ and $CAP = 1$.
- **Subtask 3 (50 pts):** Original constraints.

Examples

standard input	standard output
5 5 2 20 0 21..A.. 1 1 A 100 20	100 8 1 2 2 2 3 2 4 2 3 2 2 2 1 2 0 2
5 7 2 30 0 31... .2...3.A... 3 1 A 150 30 2 A 120 30 3 A 180 30	450 20 1 3 2 3 3 3 4 3 3 3 2 3 2 2 2 1 2 2 2 3 2 4 2 5 2 4 2 3 3 3 4 3 3 3 2 3 1 3 0 3
3 3 1 15 0 0 .T1 .T. .A. 1 1 A 100 5	0 0